The unrestricted optimum after Conjecture 20: the one-dimensional mechanism, a width theorem, and what can and cannot be proved

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Abstract

We tried to prove that the unrestricted maximum margin equals the paper's prediction, either the interference-free (IF) value $\nu^{1-s}B_p\Gamma_s(p)$ or the price–programme value $\pi_{n,b}(p)B_p\Gamma_s(p)$ of the companion note. The outcome is as follows. (1) A universal proof of the IF value is impossible. At $p = 3$, $b = 2$ we identify the mechanism behind the counterexamples exactly: a *shared-sign, one-dimensional* family whose feature field is a rotating phase times a real exponential sum. Searching this family (exact LP for the weights, outer search over m real slopes) gives new certified networks with $\gamma/\gamma_{\mathrm{IF}} = 1.063, 1.193, 1.184, 1.457, 1.617, 1.595, 2.011, 2.212$ for $n = 11, \dots, 18$. The excess grows with n. (2) We prove a *width theorem*: a slot-symmetric single-feature network whose input vectors share a sign pattern needs at least $\nu - \lfloor \nu/p \rfloor$ neurons. Every network in (1) attains this bound. The saving that drives the counterexamples is therefore at most a $1/p$ fraction of the neurons, and is zero when $p > \nu$, which covers the paper's experimental range. (3) For $p = 5, 7$ the same mechanism, run at its minimal width, loses to IF by a factor that grows with n (efficiency ≤ 0.19 for $p = 5$, $n \le 16$, and decreasing). Releasing it into the full slot-symmetric class did not help at the point we tested ($n = 10$). (4) Decoupling, the remaining step of the price–programme conjecture, is *not* automatic: at $p = 5$ a mean-zero frequency-2 field carried by the same neurons raises a feature's margin by factors 1.2–2.6. In every case we found it still stays $\ge 4\times$ below IF. So the correct universal statement is the price–programme form, and the open content is concentrated in two precise questions for $p \ge 5$.

1 Setting and notation

As in the paper and the companion note: p odd prime, $\rho = e^{2\pi i/p}$, $\theta := \pi/p$, $\nu = n + 1$, $s = \nu/b$. A *single-feature* network has output vectors at frequency 1 only; its margin is $h = \kappa_{\min}(p) \min_x \mathrm{cm}(\Phi(x))$, where $\Phi(x) = \sum_i W_i \prod_t u_{it}(x_t)\rho^{x_t}$ and $\mathrm{cm}(z) = \mathrm{Re}\, z - \cot\theta\, |\mathrm{Im}\, z|$ (companion note, Lemmas 1–2). So $h > 0$ iff every value $\Phi(x)$ lies in the open sector $|\arg z| < \theta$. A network is *slot-symmetric* if $u_{it} = u_i$ for all t; then Φ depends on x only through the count vector $N \in \mathbb{Z}^p_{\ge 0}$, $N_a = \#\{t : x_t = a\}$:

$$\Phi(N) = \sum_{i=1}^{m} W_i \prod_{a \in \mathbb{Z}_p} \big(u_i(a)\rho^a\big)^{N_a}.$$

Efficiency means $\gamma/\gamma_{\mathrm{IF}}$ at $b = 2$ (for single-feature networks at $p \ge 5$, $n \ge 3$ this is the same as the ratio of the single-feature margin to the IF single-feature margin, because $K^*(s, p) = 1$).

2 Shared-sign networks and their fields

Definition 1. A slot-symmetric network has a *shared sign pattern* $\varepsilon \in \{\pm 1\}^p$ if $\varepsilon_a u_i(a) \geq 0$ for all i, a. The *sine pattern* is $\varepsilon_a = +1$ for $0 \leq a < p/2$ and -1 for $p/2 < a < p$, with exponents $e_a := 2a$ resp. $2a - p$, a permutation of $\{0, \ldots, p-1\}$ satisfying $\varepsilon_a \rho^a = e^{i\theta e_a}$. The *one-dimensional (1-D) family* consists of the sine-pattern networks with $u_i(a) = \varepsilon_a e^{\lambda_i e_a}$ (up to scale), $\lambda_i \in \mathbb{R}$.

Lemma 2 (fields)**.** *For a sine-pattern network, $\Phi(N) = e^{i\theta\tau(N)} S(N)$ with $\tau(N) = \sum_a e_a N_a \in \{0, \ldots, (p-1)n\}$ and $S(N) = \sum_i W_i \prod_a |u_i(a)|^{N_a}$, an exponential sum with positive bases. In the 1-D family*

$$\Phi(N) = e^{i\theta\tau} \sum_{i=1}^{m} V_i e^{\lambda_i \tau}, \qquad \tau = \tau(N),$$

so the margin condition is a condition on the $L+1$ values $\tau = 0, \ldots, L$, $L = (p-1)n$, of a rotating phase times a real-exponential sum.

Proof. $u_i(a)\rho^a = |u_i(a)|\varepsilon_a \rho^a = |u_i(a)| e^{i\theta e_a}$; take the product over a with exponents N_a. □

At $p = 3$ the certified counterexample of the earlier notes ($n = 11$, width 8) is, to three digits, a 1-D network: all eight input vectors have the sign pattern $(-, +, +)$ (a translate of the sine pattern), and $\log|u_i(a)| - \log|u_i(0)| = \lambda_i \cdot (0, -1, 1)$ with ratio -1.00 ± 0.03. Re-optimising it inside the 1-D family raises its efficiency from 1.0544 to 1.0631.

3 A width theorem

Theorem 3 (width)**.** *Let p be an odd prime, $n \geq 1$, and let a slot-symmetric single-feature network with a shared sign pattern have positive margin. Then its width is*

$$m \;\geq\; \nu - \left\lfloor \frac{\nu}{p} \right\rfloor.$$

Proof. Positive margin is an open condition, so we may perturb zero entries of the u_i to small values of the prescribed sign. Hence assume all $u_i(a) \neq 0$. Put $d = (p-1)/2$. The map $a \mapsto a + d$ is a cyclic permutation of $\mathbb{Z}_p$ of odd length, so ε cannot alternate along it, and there is a symbol b with $\varepsilon_b = \varepsilon_{b+d}$. Let $a = b + d$ and consider the $n+1$ count vectors $N(k) = (n-k)\mathbf{e}_b + k\mathbf{e}_a$, $k = 0, \ldots, n$. With $r_i = u_i(a)/u_i(b) > 0$,

$$\Phi(N(k)) = C\left(e^{i(\pi-\theta)}\right)^k S(k), \qquad S(k) = \sum_i c_i r_i^k,$$

for constants $C \in \mathbb{C} \setminus \{0\}$, $c_i \in \mathbb{C}$, because $\rho^d = e^{i(\pi-\theta)}$. Positive margin gives $\arg S(k) \in I_k := (\omega_k - \theta, \omega_k + \theta)$ with $\omega_k = -\arg C - (\pi - \theta)k$. Note $\omega_k \equiv -\arg C + \theta k \pmod{\pi}$.

For $\psi \in \mathbb{R}$ let $h_\psi(k) = \mathrm{Im}(e^{-i\psi} S(k)) = \sum_i \mathrm{Im}(e^{-i\psi} c_i)\, r_i^k$. After merging equal bases this is a real exponential sum with at most m terms. By Laguerre's rule of signs it has at most $m - 1$ real zeros unless it vanishes identically, so the sequence $h_\psi(0), \ldots, h_\psi(n)$ has at most $m - 1$ sign changes. Fix a residue k_0 and take $\psi = -\arg C + \theta k_0$. The arc I_k meets the line $\psi + \pi\mathbb{Z}$ iff $\theta(k - k_0) \in (-\theta, \theta) + \pi\mathbb{Z}$, i.e. iff $k \equiv k_0 \pmod{p}$. Call these k *undetermined*. For every other k the sign of $h_\psi(k)$ is fixed by the side of the line on which I_k lies.

- If $k, k+1$ are both determined, $I_{k+1} = I_k - \pi + \theta$; an arc of length 2θ on one side, rotated by $-\pi + \theta$ and still avoiding the line, lies on the other side, so the signs differ.

- If k is undetermined and $0 < k < n$, then $k \pm 1$ are determined ($p \geq 3$), and with I_k centred on the line, I_{k-1} and I_{k+1} are the arcs of length 2θ immediately on either side of it, so $h_\psi(k-1)$ and $h_\psi(k+1)$ again have opposite signs.

So the determined values alternate in sign and there are $|D| - 1$ sign changes, where D is the set of determined k. $h_\psi \not\equiv 0$, since otherwise every $S(k)$ would lie on the line, contradicting $D \neq \emptyset$. Choosing k_0 in a least populated residue class of $\{0, \ldots, n\}$ gives $|D| = n + 1 - \lfloor \nu/p \rfloor$, hence $m - 1 \geq n - \lfloor \nu/p \rfloor$. □

Corollary 4. *(i) For $p > \nu$ a shared-sign single-feature network needs at least ν neurons, as many as the IF block. (ii) For every p, the width saving is at most $\lfloor \nu/p \rfloor$. For networks with equal amplitudes at $b = 2$, the width factor $(\nu/m)^{s-1}$ in the cost comparison with the IF block is therefore at most $\big(\nu/(\nu - \lfloor \nu/p \rfloor)\big)^{s-1} \leq \big(p/(p-1)\big)^{s-1}$: up to $(3/2)^{s-1}$ at $p = 3$, but at most $(5/4)^{s-1}$ at $p = 5$. (iii) Every certified counterexample in Table 1 attains the bound with equality.*

Remark 5. The paper's experiments use $\nu \leq 5$ and $p \geq 11$, so by (i) the mechanism behind the counterexamples has no width advantage anywhere in that range. Width is not the only lever: per unit cost, a flat sign vector has $2^{n/2}$ times the raw output of a cosine input. But numerically this lever alone never sufficed (Section 5).

4 The $p = 3$ optimum: reduction and certified values

Reduction. In the 1-D family, for fixed slopes $\lambda_1, \ldots, \lambda_m$ the cone constraints at $\tau = 0, \ldots, L$ are linear in $(\operatorname{Re} V_i, \operatorname{Im} V_i)$, and the cost amplitude of neuron i is $\sqrt{p/2}\,|V_i|\,Z(\lambda_i)^n$ with $Z(\lambda)^2 = \sum_{k<p} e^{2\lambda k}$. Maximising the margin under a weighted ℓ_1 budget is therefore a second-order cone programme (an LP after polygonal approximation of $|V_i|$). The concave $\ell_{2/\nu}$ cost is handled by reweighting; in practice the optimal amplitudes are nearly equal and uniform weights suffice. The whole problem reduces to choosing m real slopes. We search them with CMA-ES and continuation in n. All values below are then re-verified independently.

Structure of the optimum. Each term sits at the cost-optimal tilt for the stretch of τ it covers. The cost amplitudes agree to within $\sim 10\%$, consecutive terms have nearly antipodal phases ($\approx 172^\circ$ apart at $n = 11$), and the margin is active at most τ. The field is a 10^{-3}–10^{-4} residue of terms up to 10^3 times larger, kept inside the 60° sector at every count class.

n	11	12	13	14	15	16	17	18
width m (bound $\nu - \lfloor \nu/3 \rfloor$)	8 (8)	9 (9)	10 (10)	10 (10)	11 (11)	12 (12)	12 (12)	13 (13)
$\gamma/\gamma_{\mathrm{IF}}$	1.06308	1.19335	1.18360	1.45687	1.61704	1.59546	2.01105	2.21197
previously certified	1.0544	–	1.1091	1.4293	–	–	–	–

Table 1: $p = 3$, $b = 2$. All values are certified by `certify_1d.py`: weights are rounded to ten significant digits, the margin is computed in exact rational arithmetic from the logits over all count vectors (hence over all 3^n inputs), and $\gamma/\gamma_{\mathrm{IF}}$ is enclosed by interval arithmetic.

Hence $\pi_{n,2}(3)\,\nu^{s-1}$ is at least 1.06, 1.19, 1.18, 1.46, 1.62, 1.60, 2.01, 2.21 for $n = 11, \ldots, 18$. The interference-free prediction underestimates the $p = 3$ optimum by a factor that, numerically, grows by about 10% per unit of n.

5 $p \geq 5$: the mechanism loses

Gain/loss accounting. For a 1-D network with nearly equal amplitudes, $\gamma/\gamma_{\mathrm{IF}} = \zeta \cdot 2^{n/2} \cdot (\nu/m)^{s-1}$. Here $2^{n/2}$ is the raw advantage of flat sign vectors over cosines, $(\nu/m)^{s-1}$ the width saving (capped by Theorem 3), and ζ the fraction of the mid-range envelope that survives cancellation. The available gain per unit of n is $\frac{1}{2}\log\frac{2p}{p-1}$: 0.55 at $p=3$, 0.46 at $p=5$, 0.42 at $p=7$. Numerically, $\log\zeta/n \approx -0.48$ at $p=3$ ($n=18$) and ≈ -0.575 at $p=5$ (stable over $n=13$–17). That leaves net rates of about $+0.07$ and -0.12 per unit of n, consistent with the observed growth at $p=3$ and decline at $p=5$.

n	6	7	8	9	10	11	12	13	14	15	16
$p=5$	.192	.179	.166	.147	.132	.127	.126	.120	.108	.097	.093
width (bound)	6 (6)	7 (7)	8 (8)	9 (8)	10 (9)	10 (10)	11 (11)	12 (12)	13 (12)	14 (13)	14 (14)
$p=7$	.126	.107	.089	.076	.066	.063	.058	.049	.044	.035	–

Table 2: Best 1-D efficiency $\gamma/\gamma_{\mathrm{IF}}$ found (slope search with exact LP, continuation in n with insertion and deletion moves and fresh global candidates at every n).

The 1-D restriction is not the bottleneck. At $p=5$, $n=10$, released into the full slot-symmetric single-feature class (arbitrary input vectors, `cone.py`), the 1-D optimum 0.1318 is a local optimum (trial with no noise: 0.1318; with noise: lower). Mixing in positive-input neurons, which keeps the field a function of τ but breaks the shared sign pattern, lowered the efficiency at $p=3$, $n=11$ ($1.063 \to 0.82$).

6 Cross-feature compensation: decoupling is not automatic

At $p \geq 5$ several frequencies coexist, and the margin condition at x is that the vector $(\Phi_l(x))_l$ lies in a polyhedron, not that each $\Phi_l(x)$ lies in its own sector. Concretely, at $p=5$ a frequency-2 value $\Phi_2 \approx 0.25|\Phi_1|e^{-i18^\circ}$ makes Φ_1 admissible at $\arg\Phi_1 = 60^\circ$, the $p=3$ sector. The same neurons can produce such a Φ_2 almost for free: the field is a 10^{-3} residue of much larger terms, so a frequency-2 output component with $\sim 10^{-3}$ relative weight, costing $\sim 10^{-6}$ of the budget, gives a Φ_2 of the right size. It is limited only by geometry: its phase follows a different linear functional of the counts.

We tested this with an exact LP over output vectors spanning frequencies 1 and 2, at the slopes of the best 1-D networks, on all count classes and all four shifts δ. Measured gains over frequency-1 outputs at the same neurons: $\times 1.17$ ($n=8$), $\times 1.36$ ($n=12$), up to $\times 2.6$ at non-optimal widths. With a CMA search over slopes for the joint problem: 0.197 ($n=8$) and 0.173 ($n=12$). The frequency-2 output carries 0.2–3% of the output energy. So the decoupling half of the price–programme conjecture (companion note, Conjecture 20′) cannot be proved feature by feature: one feature can live outside its own sector where another, interference-only, field compensates. In every configuration we tried, the compensated feature still stays $\geq 4\times$ below IF.

7 The number of features of the unrestricted optimum

The paper's target is the *count*: a frequency j is a feature of θ if $A_j(\theta) = |\beta_j(\theta)| > 0$, and the prediction is that every globally margin-optimal network uses $K^*(s,p) = |\operatorname{spt} y^\star|$ features, with amplitudes proportional to a maximiser $y^\star$ of the programme $\Gamma_s(p)$. This is weaker than the value prediction, and it survives the counterexamples.

Proposition 6 (what is unconditional)**.** *Let $\theta^\star$ be a global maximiser of the $L_{2,b}$-margin.*

(a) If $p = 3$, then $\theta^\star$ uses exactly $K^*(s, 3) = 1$ feature, for every n and b.

(b) If $\theta^\star$ is interference-free, its amplitudes are proportional to a maximiser of $\Gamma_s(p)$, so it uses exactly $|\operatorname{spt} y^\star|$ features (paper, Theorem 8(1)).

(c) If $b_0 \le b \le \nu$ (interval of the earlier notes, Part II), then $\theta^\star$ is interference-free and uses all $F = K^*(s, p)$ frequencies with equal amplitudes.

(d) For every p, n, b: write $\eta(\theta) := \|A(\theta)\|_{1/s}/(B_p\|\theta\|_{2,b}^\nu) \le 1$ for the clean efficiency and $y = A/\|A\|_{1/s}$ for the clean allocation of $\theta^\star$. Then $\eta(\theta^\star) \ge \pi_{n,b}(p)$ and $\min_{\delta\neq 0}\sum_j y_j\kappa_j(\delta) \ge \frac{\pi_{n,b}(p)}{\eta(\theta^\star)}\,\Gamma_s(p)$. In particular, if $\eta(\theta^\star) = \pi_{n,b}(p)$, then y is a programme maximiser and $\theta^\star$ uses exactly $|\operatorname{spt} y|$ features.

Proof. (a) $F = \{1\}$, and $A_1 > 0$ by the paper's Lemma 2 because the margin is positive. (b), (c) are the cited results. (d) By the price–programme theorem of the companion note, $h(\theta^\star) = \gamma_b^*\|\theta^\star\|^\nu \ge \pi B_p\Gamma_s\|\theta^\star\|^\nu$. By the paper's Lemma 2, $h \le \min_\delta \sum_j A_j\kappa_j(\delta) = \|A\|_{1/s}\min_\delta\sum_j y_j\kappa_j(\delta) \le \|A\|_{1/s}\Gamma_s$. Divide by $B_p\|\theta^\star\|^\nu$. □

Proposition 7 (decoupling suffices; the price is irrelevant)**.** *Suppose a global maximiser $\theta^\star$ satisfies the decoupling inequality*

$$h(\theta^\star) \;\le\; \pi_{n,b}(p)\,B_p\,\min_{\delta\neq 0}\sum_j P_j^s\kappa_j(\delta) \qquad \text{for some } P_j \ge 0,\quad \sum_j P_j \le \|\theta^\star\|_{2,b}^b.$$

Then $(P_j^s/\|\theta^\star\|_{2,b}^\nu)_j$ is a maximiser of $\Gamma_s(p)$. If moreover the features of $\theta^\star$ are exactly the frequencies it pays for ($A_j > 0 \iff P_j > 0$), then $\theta^\star$ uses exactly K^* features.

Proof. With $z_j = P_j^s/\|\theta^\star\|^\nu$ we have $\sum_j z_j^{1/s} \le 1$, and

$$\pi B_p\Gamma_s\|\theta^\star\|^\nu \le h(\theta^\star) \le \pi B_p\|\theta^\star\|^\nu\min_\delta\sum_j z_j\kappa_j(\delta).$$

Hence z attains Γ_s. □

The price enters only as a common factor, because relabelling $a \mapsto ja$ makes it the same for every frequency and scaling makes it homogeneous of degree s in the budget. So the count needs *only* decoupling. Neither $\pi = \nu^{1-s}$ nor interference-freeness is needed.

Theorem 8 (exact count in the dense regime; lower bound in general)**.** *For $S \subseteq F$ let $\Gamma_s^{(S)}(p)$ be the programme restricted to allocations supported on S, and $K_{\mathrm{low}}(s, p, \nu) := \min\{|S| : \Gamma_s^{(S)}(p) \ge \nu^{1-s}\Gamma_s(p)\}$. Every global maximiser of the $L_{2,b}$-margin, of any width and with interference allowed, uses at least K_{low} features. In particular, if*

$$\nu^{s-1}\,\Gamma_s^{(F\setminus\{j\})}(p) < \Gamma_s(p) \qquad (\textit{independent of } j \textit{ by relabelling}),$$

every global maximiser uses all $F = (p-1)/2$ frequencies. This is the exact count $K^*(s, p) = F$ in the dense regime. The condition holds for $s \le 1.331, 1.285, 1.257$ ($p = 5$; $\nu = 3, 4, 5$), $s \le 1.154, 1.130, 1.115$ ($p = 7$), $s \le 1.079, 1.066, 1.058$ ($p = 11$), $s \le 1.064, 1.053, 1.047$ ($p = 13$). In every case computed this is 2.5 to 5 times the range $s - 1 \le 1/(F(2\nu + 1) - 2)$ of Part II.

Proof. If the features of $\theta^\star$ lie in S, Proposition 3 of the paper restricted to S and the single-neuron budget give $\gamma(\theta^\star) \le B_p\Gamma_s^{(S)}$, while $\gamma(\theta^\star) = \gamma_b^* \ge \nu^{1-s}B_p\Gamma_s$ (Theorem 8). Relabelling $a \mapsto ka$ maps allocations on S to allocations on kS with the same value, and acts transitively on F. □

Why the upper bound is the hard half. Let q be KKT multipliers of a maximiser on its active constraints (x,δ). Stationarity in the output vector makes neuron i's frequency-j output proportional to $\sum_x g_i(x)\rho^{-jy(x)}\overline{Q_j(x)}$, with $g_i(x)=\prod_t u_{it}(x_t)$ and $Q_j(x)=\sum_\delta q(x,\delta)(1-\rho^{-j\delta})$. If q is uniform in x, as for any interference-free maximiser (average q over translations), this gives $\beta_{ij} \propto Q_j\,|\sum_x g_i(x)\rho^{-jy(x)}|^2$ with $\operatorname{Re} Q_j = \sum_\delta q(\delta)\kappa_j(\delta) > 0$. Then every frequency at which some neuron's input product has a non-zero clean projection is a feature. So "no more than K^* features" is equivalent to frequency purity of the maximiser's input products at the non-programme frequencies. Interference-free maximisers have it (paper, Theorem 8(2)). A maximiser built from sign-vector inputs, like the $p=3$ networks, would generically carry every frequency. The upper bound is therefore not separable from the question of which input structure is optimal at $p\ge 5$.

Theorem 9 (the count is $\Theta(\log p)$ for every near-optimal network)**.** *Let $2\le b<\nu$, $s=\nu/b$, $r=1/s$, and for a non-zero amplitude vector A let*

$$K_r(A) := \frac{\big(\sum_j A_j^r\big)^{1/(1-r)}}{\big(\sum_j A_j\big)^{r/(1-r)}} \;\le\; |\operatorname{spt} A|,$$

the participation number of A (it equals K for K equal non-zero entries). Let θ be any product network, of any width and with interference allowed, with $\gamma(\theta)\ge \lambda\,\nu^{1-s}B_p\Gamma_s(p)$ for some $\lambda\in(0,1]$. Every global maximiser has $\lambda\ge \pi_{n,b}(p)\nu^{s-1}\ge 1$. Then

$$K_r(A(\theta)) \;\le\; \nu\Big(\frac{p}{(p-1)\lambda\,\Gamma_s(p)}\Big)^{\frac{1}{s-1}} \;\le\; \nu\Big(\frac{2p}{(p-1)\lambda}\Big)^{\frac{1}{s-1}}\lceil 8\ln p\rceil.$$

Moreover, for every $M\ge 1$ the M largest amplitudes carry at least a fraction $1-(K_r/M)^{1-r}$ of $\sum_j A_j$. Together with the paper's Corollary 12, which shows that the support of such a network has at least $c_{\nu,b,\lambda}\ln p$ frequencies, the feature count of every margin-optimal network is $\Theta_{\nu,b}(\log p)$. This is the order of $K^*(s,p)$ (paper, Theorem 11), for every odd prime, unconditionally.

Proof. Averaging the paper's Lemma 2 over $\delta\neq 0$ and using $\sum_{\delta\neq 0}\kappa_j(\delta)=p$ gives $h\le \frac{p}{p-1}\sum_j A_j$. The single-neuron budget gives $\sum_j A_j^r\le \sum_{i,j}|\beta_{ij}|^r\le C_bP$ with $P=\|\theta\|_{2,b}^b$. Hence, with $h=\gamma P^s$ and $sr=1$,

$$K_r(A)\le \frac{(C_bP)^{1/(1-r)}}{\big(\frac{p-1}{p}\gamma P^s\big)^{r/(1-r)}} = \Big(\frac{p\,C_b^{1/r}}{(p-1)\gamma}\Big)^{\frac{r}{1-r}} = \Big(\frac{p\,B_p}{(p-1)\gamma}\Big)^{\frac{1}{s-1}},$$

because $C_b^{1/r}=C_b^{\nu/b}=B_p$ and $r/(1-r)=1/(s-1)$. Insert $\gamma\ge\lambda\nu^{1-s}B_p\Gamma_s$, then $\Gamma_s\ge\frac12\lceil 8\ln p\rceil^{1-s}$ (paper, Theorem 10). The bound $\lambda\ge\pi\nu^{s-1}$ for maximisers is the price–programme theorem. For the tail statement, normalise $\sum_j A_j=1$ and sort; the amplitudes beyond the M-th are at most $1/M$ each, so their mass t satisfies $tM^{1-r}\le\sum_{j>M}A_j^r\le K_r^{1-r}$. □

Corollary 10 ($n=2$, every prime $p\ge 5$)**.** *For $n=2$ and $2\le b\le 3$, no single-frequency network (inputs in $\operatorname{span}\{\mathbf{1},\cos,\sin\}$ of one frequency, any width) beats the interference-free single-feature value at any prime $p\ge 5$.*

Proof. For $n=2$ the robust constant of the companion note's Theorem 14 is $k^*(2,s)=\sup\,(\operatorname{Re}A - 3^{1-s}(\sum|\beta_i|^{1/s})^s)/(R/\sqrt{2})$ with $R^2=|B|^2+|C|^2+|D|^2$, where B,C,D are the three sign-flipped sums. The four-moment inequality of the earlier notes (Part I, Theorem A: $|A|-R/(2\sqrt{3})\le 3^{1-s}(\sum|\beta_i|^{1/s})^s$ for $1\le s\le 3/2$, valid for arbitrary unit phases) gives $k^*(2,s)\le\sqrt{2}/(2\sqrt{3})=1/\sqrt{6}<0.41<k_5=0.946\le k_p$. The companion note's Theorem 14 then applies. □

Lemma 11 (slack identity)**.** *For every logit table (any network, any p, n, any inputs), with margin h and slack $\sigma(x,\delta) := f(x)[y(x)] - f(x)[y(x)-\delta] - h \geq 0$,*

$$\Phi_j(x) = \tfrac{2}{p}\Big(h - \sum_{\delta\neq 0} \sigma(x,\delta)\rho^{j\delta}\Big) \qquad (j \in F).$$

Consequently every sign-flipped coefficient satisfies $|\beta_{j,T}| \leq \frac{2}{p}S$ with $S := \mathbb{E}_x \sum_\delta \sigma = p\sum_{j\in F} \operatorname{Re}\beta_j - (p-1)h$. The same slack S bounds the interference of every *frequency.*

Proof. $\hat{f}_x(j) = \sum_\delta f(x)[y-\delta]\rho^{-j(y-\delta)} = -\rho^{-jy}\sum_{\delta\neq 0}(h+\sigma(x,\delta))\rho^{j\delta}$ because $\sum_\delta \rho^{j\delta} = 0$, and $-\sum_{\delta\neq 0}\rho^{j\delta} = 1$. Then $\beta_{j,T} = \overline{\mathbb{E}_x \Phi_j \chi}$ for a unimodular character χ, and $|\mathbb{E}[\sigma\chi]| \leq \mathbb{E}\sigma$. For S, use $\mathbb{E}_x\Phi_j = \overline{\beta_j}$ and that $\sum_{j\in\mathbb{Z}_p}\hat{\bar{\sigma}}(j) = p\,\bar{\sigma}(0) = 0$. □

Proposition 12 ($n = 2$, any inputs)**.** *For $n = 2$, $2 \leq b \leq 3$, every prime p and every network, $h \leq \frac{p}{p-1}\big(\sum_j \operatorname{Re}\beta_j - \max_j R_j/(2\sqrt{3})\big)$, where $R_j^2 = \sum_{T\neq\emptyset}|\beta_{j,T}|^2$. For single-feature networks (any inputs) this gives $\gamma \leq \frac{p}{p-1}\nu^{1-s}B_p$, i.e. $\pi_{2,b}(p) \leq \nu^{1-s}\cdot\frac{p}{(p-1)\kappa_{\min}(p)}$. This is Part I's sharp result at $p = 3$, but it is weaker than the IF value at $p \geq 5$.*

Proof. Lemma 11 with three non-empty T gives $R_j \leq 2\sqrt{3}(\sum_{j'}\operatorname{Re}\beta_{j'} - \frac{p-1}{p}h)$. For one feature, apply Part I's Theorem A ($|A| - R/(2\sqrt{3}) \leq 3^{1-s}(\sum|\beta_i|^{1/s})^s$; Lemma 6 of the paper supplies the unit phases for arbitrary inputs) and the single-neuron budget. □

Two losses separate Proposition 12 from the exact answer at $p \geq 5$, and both are the multi-frequency obstacle. (i) A single feature's gaps at its non-binding offsets carry structural slack, which uses up S without any interference. (ii) With several frequencies, one slack budget bounds each R_j but not $\sum_j R_j$, so only one frequency's interference is charged. Part II of the earlier notes overcomes (ii) only near $b = \nu$, by mixing the chain matrices of all frequencies.

The general $n = 2$ case (inputs with other frequencies) is also where the L^2 route stops. Input energy ε outside ± 1 costs only $O(\varepsilon)$ of clean amplitude, but it can create non-circular (partner-mode) interference of size $O(\sqrt{\varepsilon})$. So every L^2/moment bound fails in a thin regime next to the interference-free point, and a pointwise argument, like Part I's at $p = 3$, is required.

Numerical tests of the count with general networks. (i) *Width.* For slot-symmetric networks with arbitrary input vectors (any sign patterns) and outputs on all frequencies, we searched for the smallest width with positive margin (elimination from feasible starts, exact LP for the outputs). It equals $\nu - \lfloor \nu/p \rfloor$ in every case tested: $p = 3$, $n = 3, 4, 5, 6, 8$ (widths $3, 4, 4, 5, 6$), and $p = 5$, $n = 4, 5, 6, 7$ (widths $4, 5, 6, 7$), with one output frequency or all of them. So mixing frequencies does not save neurons, and Theorem 3 appears to hold for all slot-symmetric networks. (ii) *Allocation.* At $(p, n, b) = (7, 3, 2)$ the programme has $K^* = 2$ with unequal shares $(0.328, 0, 0.182)$. Global search (inputs by CMA-ES from interference-free blocks on every frequency subset plus perturbations, outputs by exact conic programme with ℓ_r reweighting) gave: one frequency 0.801; any two frequencies 1.00000 with amplitude shares equal to the programme's up to relabelling, unchanged under perturbation; all three frequencies 0.827. At $(5, 2, 2)$ ($K^* = 2$), random-start search reached 0.755 (width 4, one feature), 0.821, 0.941, 0.970 (widths 5–7, two features with near-equal shares), and never exceeded the K^*-feature block.

Where a proof stops. Combining (a)–(c): the count prediction holds for every global optimiser, *except possibly* when $p \geq 5$, $b < b_0$ and the optimiser carries interference. None of the known counterexamples is in that set, since they all have $p = 3$. Closing the remaining case requires a price-sharp upper bound, i.e. Conjecture 20′ or an equivalent. The only unconditional upper bound, Theorem 4 of the paper, is too weak: it loses the factor $\pi\nu^{s-1}$, which swamps the differences between allocations (removing one feature changes Γ_s by a factor $(1 - w_j)^s$ only, paper Lemma 19). The compensation of Section 6 is the precise danger. It uses frequencies whose clean amplitude is tiny but non-zero and that the programme does not pay for. An optimiser built that way would *over*count, so the strict count statement at $p \geq 5$ is exactly as hard as excluding compensation at the optimum. In every experiment at $p = 5, 7$, compensated networks stay $\geq 4\times$ below the K^*-feature IF block. Section 8 proves the count for a class that carries interference, and isolates what is left.

8 A Lipschitz form of Lemma 7 and a transport reduction

Lemma 13 (robust interference-free bound, every n)**.** *Let $0 < r \leq 1$, $s = 1/r$, $\beta_1, \ldots, \beta_N \in \mathbb{C}$, $\xi_l \in \mathbb{T}^n$, $A = \sum_l \beta_l$, $\beta_T = \sum_l \beta_l \prod_{t \in T} \xi_{lt}$, $Q = \sum_l |\beta_l|^r$, and let $\ell := \max |\beta_T|$ over the non-empty* intervals $T = \{t' + 1, \ldots, t\} \subseteq [n]$. *Then*

$$\textit{(i)}\ \ |A|^r \leq \nu^{r-1} Q + \|L\|_{\mathrm{op}}^r, \qquad \textit{(ii)}\ \ |A| \leq \nu^{1-s} Q^s + K_{\nu,s}\, \ell, \quad K_{\nu,s} := \tfrac{\nu! - 1}{\nu}\, \nu^{(s-1)(\nu-1)},$$

where L is the strictly lower part of the chain matrix. With $\ell = 0$ both are the paper's Lemma 7.

Proof. Let $H = \sum_l \beta_l m_l m_l^*$ with the chain vectors m_l of the paper's Lemma (chain). Its diagonal is A, its strictly lower entries are the interval sums β_T, so $H = H_0 + L$ with $H_0 = AI + U$ upper triangular and $|L_{tt'}| \leq \ell$. Rotfel'd's inequality for $0 < r \leq 1$ gives $\sum_k \sigma_k(H)^r \leq \sum_l \|\beta_l m_l m_l^*\|_{S_r}^r = \nu^r Q$. (i) All eigenvalues of H_0 equal A, so Weyl's majorant theorem gives $\sum_k \sigma_k(H_0)^r \geq \nu |A|^r$. Weyl's perturbation bound $\sigma_k(H_0) \leq \sigma_k(H) + \|L\|$ and subadditivity of $t \mapsto t^r$ give $\nu|A|^r \leq \nu^r Q + \nu \|L\|^r$. (ii) Every non-identity permutation has a factor below the diagonal, and every entry of H has modulus at most $S_1 := \sum_l |\beta_l| \leq Q^s$. So $|\det H - A^\nu| \leq (\nu! - 1)\, \ell\, S_1^{\nu-1}$. By AM–GM on σ_k^r, $|\det H| \leq (\nu^{-1} \sum_k \sigma_k^r)^{\nu/r} \leq a^\nu$ with $a := \nu^{1-s} Q^s$. Hence $|A|^\nu \leq a^\nu + (\nu! - 1) \ell Q^{s(\nu-1)}$, and $(a + \varepsilon/(\nu a^{\nu-1}))^\nu \geq a^\nu + \varepsilon$ gives (ii), since $Q^{s(\nu-1)}/a^{\nu-1} = \nu^{(s-1)(\nu-1)}$. □

By the paper's Lemma 6 the lemma applies to the neurons of any network at any frequency j, with $A = \beta_j$ and $\ell = \ell_j := \max_{T \text{ interval}} |\beta_{j,T}|$.

Corollary 14 (single-frequency networks, every n, all large p)**.** *The constant of the companion note's Theorem 14 satisfies $k^*(n, s) \leq \sqrt{2}\, K_{\nu,s} < \infty$ for every n and s. Since $k_p = \sqrt{\cot^2(\pi/p) - 1} \geq p/\pi - 1$ for $p \geq 5$, no single-frequency network beats the interference-free single-feature value at any prime $p \geq \pi(1 + \sqrt{2} K_{\nu,s})$. (For $n = 2$, Corollary 10 already gives every $p \geq 5$.)*

Proof. $\mathrm{Re}\, A - \nu^{1-s} Q^s \leq K_{\nu,s} \ell \leq K_{\nu,s} \sqrt{E} = \sqrt{2} K_{\nu,s} \sqrt{E/2}$. For k_p: $\cot x \geq 1/x - x/2$ on $(0, \pi/5]$ (the difference vanishes at 0 and has derivative $1/x^2 + 1/2 - \csc^2 x \geq 0$ there), so $\cot^2 x - 1 \geq x^{-2} - 2 \geq (x^{-1} - 1)^2$ for $x \leq 2/3$. □

The constant is crude ($K_{3,3/2} = 5$, $K_{4,2} = 368$), so the threshold is large, but it is finite and explicit for every n.

Definition 15. A network is *frequency-pure without input DC* (class $\mathcal{C}$) if every neuron i has a frequency $j_i \in F$ with $u_{it} \in \mathrm{span}\{\cos(2\pi j_i \cdot /p), \sin(2\pi j_i \cdot /p)\}$ for all t and $w_i \in \mathrm{span}\{\mathbf{1}, \cos(2\pi j_i \cdot /p), \sin(2\pi j_i \cdot /p)\}$. Neurons of different frequencies may coexist in any numbers. Every interference-free block of the paper is in $\mathcal{C}$.

For $\theta \in \mathcal{C}$, expanding $u(a)\rho^{la} = p^{-1}(\hat{u}(-l) + \hat{u}(l)\rho^{2la})$ gives $\Phi_l(x) = \sum_{T\subseteq[n]} a_{l,T}\,\rho^{2ls_T(x)}$ with $s_T(x) = \sum_{t\in T} x_t$, $a_{l,\emptyset} = \overline{\beta_l}$ and $|a_{l,T}| = |\beta_{l,T}|$ (companion note, Lemma 11). For $T \neq \emptyset$ define the *interference profile*

$$\varphi_T(w) := \sum_{l\in F} \operatorname{Re}\left(a_{l,T}\rho^{lw}\right), \qquad w \in \mathbb{Z}_p,$$

a mean-zero real function on $\mathbb{Z}_p$ whose Fourier coefficients are the sign-flipped coefficients of all features at once. Write $m_\delta := \mathbb{E}_x g(x,\delta) - h \geq 0$ for the mean slack and $\bar{m}_\delta = \frac{1}{2}(m_\delta + m_{-\delta})$.

Lemma 16 (interference profiles are slack potentials)**.** *Let $\theta \in \mathcal{C}$, p any odd prime. For every non-empty T, every $w \in \mathbb{Z}_p$ and every $\delta \neq 0$,*

$$\varphi_T(w) - \varphi_T(w-\delta) \;\geq\; -m_\delta.$$

Consequently $|\beta_{l,T}|\,|1-\rho^{l\delta}| \leq 2m_\delta$ for all l, T, δ, i.e. $|\beta_{l,T}| \leq \sqrt{2}\,\min(m_\delta, m_{-\delta})/\sqrt{\kappa_l(\delta)}$.

Proof. With $c_l(\delta) = 1-\rho^{-l\delta}$, $\operatorname{Re}(a_{l,T}c_l(\delta)\rho^{2ls_T}) = \operatorname{Re}(a_{l,T}\rho^{l\cdot 2s_T}) - \operatorname{Re}(a_{l,T}\rho^{l(2s_T-\delta)})$, so by the companion note's Lemma 2

$$g(x,\delta) = \mathbb{E}_x g(\cdot,\delta) + \sum_{T\neq\emptyset} \big(\varphi_T(2s_T(x)) - \varphi_T(2s_T(x)-\delta)\big).$$

Fix $T_0 \neq \emptyset$ and condition on $s_{T_0}(x) = z$. For $T \neq T_0$ the forms s_T, s_{T_0} are linearly independent over $\mathbb{Z}_p$, so s_T is conditionally uniform and φ_T averages to 0. Hence $h \leq \mathbb{E}[g \mid s_{T_0} = z] = h + m_\delta + \varphi_{T_0}(2z) - \varphi_{T_0}(2z-\delta)$; put $w = 2z$ (p odd). For the consequence, $w \mapsto m_\delta + \varphi_T(w) - \varphi_T(w-\delta)$ is non-negative with mean m_δ, and its coefficient at ρ^{lw} is $\frac{1}{2}a_{l,T}c_l(\delta)$, with $|c_l(\delta)| = \sqrt{2\kappa_l(\delta)}$. □

So the pointwise margin constraints, projected onto each T, say exactly that $\varphi_T(w') - \varphi_T(w) \leq m_{w-w'}$: the interference profile is a potential for the slack, i.e. 1-Lipschitz for the path quasi-metric D_m that the mean slacks induce on the Cayley graph of $\mathbb{Z}_p$. By Kantorovich–Rubinstein duality, the largest coefficient a feature l can carry at T is a transport cost: $\max|\beta_{l,T}| = \frac{2}{p}\max_\phi W_{D_m}(\nu^+_{l,\phi}, \nu^-_{l,\phi})$, $\nu_{l,\phi}(w) = \cos(2\pi lw/p + \phi)$. No circularity is used, so this holds at $p = 3$ as well.

Theorem 17 (decoupling when few features are bought cheaply)**.** *Let $\theta \in \mathcal{C}$, let $a_l := \nu^{1-s}Q_l^s$ with $Q_l = \sum_i |\beta_{il}|^r$ be the interference-free price of the amplitudes θ pays for, let $\delta^\star$ minimise $\sum_l a_l\kappa_l(\delta)$, and let $L' = \{l : |\beta_l| > a_l\}$ be the frequencies bought* below *the interference-free price. If*

$$\sum_{l\in L'} \kappa_l(\delta^\star) \;\leq\; \frac{2}{(2^n-1)\,k^*(n,s)^2} \qquad (= 4 \text{ for } n = 2\,),$$

then $\gamma(\theta) \leq \nu^{1-s}B_p\Gamma_s(p)$, and if the inequality is strict, equality forces θ to be interference-free. For $n = 2$ the condition holds whenever $|L'| \leq 2$, because $\kappa_l < 2$. Hence, for $n = 2$, every odd prime p and every $b \in [2,3]$: among networks of $\mathcal{C}$ in which at most two frequencies are bought below the interference-free price (any width, any interference elsewhere), the maximum is the interference-free value, and every maximiser is interference-free and uses exactly $|\operatorname{spt} y^\star| = K^*(s,p)$ features.

Proof. Averaging $g(\cdot,\pm\delta)$ over x gives, for every δ, $h = \sum_l \operatorname{Re}\beta_l\,\kappa_l(\delta) - \bar{m}_\delta$. By the companion note's Theorem 14 (whose constant is finite by Corollary 14), $|\beta_l| - a_l \leq k^*\sqrt{E_l/2}$ with $E_l = \sum_{T\neq\emptyset}|\beta_{l,T}|^2$. For $n = 2$ take $k^* = 1/\sqrt{6}$ (Part I, Theorem A). Fix T. By the proof of Lemma 16, $D_T(w) := \varphi_T(w) - \varphi_T(w-\delta^\star)$ has mean zero and values in $[-m_{\delta^\star}, m_{-\delta^\star}]$ (the upper bound is the constraint at $(w-\delta^\star, -\delta^\star)$). So $\frac{1}{p}\sum_w D_T^2 \leq m_{\delta^\star}m_{-\delta^\star} \leq \bar{m}^2$ (Bhatia–Davis), and Parseval, with

coefficients $a_{l,T}c_l(\delta^\star)$ and $|c_l|^2 = 2\kappa_l$, gives $\sum_l |\beta_{l,T}|^2\kappa_l(\delta^\star) \le \bar{m}^2$. Summing over T: $\sum_l E_l\kappa_l(\delta^\star) \le (2^n-1)\bar{m}^2$. By Cauchy–Schwarz,

$$h \le \sum_l a_l\kappa_l(\delta^\star) + \frac{k^*}{\sqrt{2}}\sum_{l\in L'}\sqrt{E_l}\,\kappa_l(\delta^\star) - \bar{m} \le \sum_l a_l\kappa_l(\delta^\star) + \bar{m}\Big(k^*\sqrt{\tfrac{2^n-1}{2}\textstyle\sum_{l\in L'}\kappa_l(\delta^\star)} - 1\Big).$$

Under the condition the bracket is ≤ 0, so $h \le \nu^{1-s}\min_\delta \sum_l Q_l^s\kappa_l(\delta) \le \nu^{1-s}(C_bP)^s\Gamma_s$ by the single-neuron budget $\sum_l Q_l \le C_bP$. If the bracket is < 0 and equality holds, then $\bar{m} = 0$, and Lemma 16 gives $\beta_{l,T} = 0$ for all l, T. So the maximiser is interference-free, and the paper's Theorem 8 gives its amplitudes and its count. □

Theorem 17 is the first count result at $p \ge 5$ outside the dense regime for a class of networks that carry interference. Any number of features may carry interference, and up to two of them (for $n = 2$) may use it to buy themselves below the interference-free price. They still cannot beat the interference-free value, because the slack their interference forces at $\delta^\star$ is at least as large as what they gain. What it does not cover is *collective* undercutting: many features each bought a little below price, with phases arranged so that their profiles nearly cancel. The proof charges these through Cauchy–Schwarz over L', which loses $\sqrt{|L'|}$. A proof of the general count must replace this step by an estimate that sees the cancellation.

The count fails only through cheap features. The next proposition holds for every n, p, b and every network. It says that the count can fail only if the maximiser buys some frequency *below* its interference-free price, so everything reduces to excluding cheap features.

Proposition 18 (no cheap features ⇒ exact count)**.** *Let $\theta^\star$ be a global maximiser of the $L_{2,b}$-margin, $a_l = \nu^{1-s}Q_l^s$, $Q_l = \sum_i |\beta_{il}|^r$. If $|\beta_l| \le a_l$ for every l, then $(Q_l^s)_l$ is proportional to a maximiser $y^\star$ of $\Gamma_s(p)$, $|\beta_l| = a_l$ for every l, and $\theta^\star$ uses exactly $|\operatorname{spt} y^\star|$ features.*

Proof. $h \le \min_\delta \sum_l \operatorname{Re}\beta_l\kappa_l(\delta) \le \nu^{1-s}\min_\delta\sum_l Q_l^s\kappa_l(\delta) \le \nu^{1-s}(\sum_l Q_l)^s\Gamma_s \le \nu^{1-s}(C_bP)^s\Gamma_s$, and the left side is at least the right side because interference-free networks attain it (paper, Theorem 8). So every inequality is an equality. The third makes (Q_l^s) proportional to a programme maximiser. The first two give $|\beta_l| = a_l$ wherever $\kappa_l(\delta) > 0$, i.e. everywhere, so $\beta_l \ne 0 \iff Q_l > 0$. □

Proposition 18 does not need the maximiser to be interference-free. At $p = 3$ the counterexamples buy their single feature below price, consistent with the proposition.

Theorem 19 (Riesz products: cheap features need arithmetic coincidences, $n = 2$)**.** *Let $n = 2$, $2 \le b \le 3$, p any odd prime, and θ any network (inputs and outputs arbitrary). Write $\Phi_l(x) = \sum_{k\in\mathbb{Z}_p^2}\phi_l(k)\rho^{k\cdot x}$; the sign-flipped coefficient sits at the character $\chi_{l,T} := 2l\,\mathbf{1}_T$ with $|\phi_l(\chi_{l,T})| = |\beta_{l,T}|$ (the computation of the companion note's Lemma 11 uses no property of the inputs). Let $\hat{g}_\delta(k)$ be the Fourier coefficients of $x \mapsto g(x,\delta)$, let $\delta^\star$ minimise $\sum_l a_l\kappa_l(\delta)$, and let $L' = \{l : |\beta_l| > a_l\}$. Suppose there are T_l ($l \in L'$) with $|\beta_{l,T_l}| = \max_T|\beta_{l,T}|$ such that $\Lambda = \{\chi_{l,T_l}\}_{l\in L'}$ is*

- *(D)* dissociated*: the $3^{|L'|}$ sums $\sum_l \sigma_l\chi_{l,T_l}$, $\sigma \in \{0,\pm1\}^{L'}$, are pairwise distinct (no relation with coefficients in $\{0,\pm1,\pm2\}$); and*
- *(I)* isolated *at $\pm\delta^\star$: $\hat{g}_{\pm\delta^\star}$ vanishes on the sums $\sum_l\sigma_l\chi_{l,T_l}$ with $|\operatorname{spt}\sigma| \ge 2$, and at $\pm\chi_{l,T_l}$ only feature l's own sign-flipped mode contributes.*

Then $\gamma(\theta) \le 3^{1-s}B_p\Gamma_s(p)$, strictly if $L' \ne \emptyset$. Hence every global maximiser either has no cheap feature, and then uses exactly $K^*(s,p)$ features (Proposition 18), or its cheap features violate (D) or (I).

Proof. For any $\nu \geq 0$ with $\mathbb{E}\nu = 1$, $h \leq \mathbb{E}_x[\nu(x)g(x,\delta)]$. Take $\nu = \prod_{l\in L'} \big(1-\cos(2\pi\langle \chi_{l,T_l}, x\rangle/p+\psi_l)\big)$ with $\psi_l = \arg \hat{g}_\delta(\chi_{l,T_l})$. By (D) every character of ν has a unique representation, so $\hat{\nu}(0) = 1$ and $\hat{\nu}(\pm\chi_{l,T_l}) = -\frac{1}{2}e^{\pm i\psi_l}$, and $\hat{\nu}$ is supported on $\{\sum \sigma_l \chi_{l,T_l}\}$. By (I),

$$h \leq \hat{g}_\delta(0) - \sum_{l\in L'} |\hat{g}_\delta(\chi_{l,T_l})| = \mathbb{E}_x g(\cdot,\delta) - \sum_{l\in L'} |\beta_{l,T_l}|\sqrt{\kappa_l(\delta)/2},$$

because $|\hat{g}_\delta(\chi_{l,T_l})| = \frac{1}{2}|\beta_{l,T_l}||1-\rho^{-l\delta}|$. Average this over $\delta = \pm\delta^\star$ and use Part I's Theorem A for $l \in L'$ ($|\beta_l| - a_l \leq R_l/(2\sqrt{3}) \leq \frac{1}{2}|\beta_{l,T_l}|$; Lemma 6 of the paper makes it valid for any inputs):

$$h \leq \sum_l a_l\kappa_l(\delta^\star) + \sum_{l\in L'} |\beta_{l,T_l}|\Big(\tfrac{1}{2}\kappa_l(\delta^\star) - \sqrt{\kappa_l(\delta^\star)/2}\Big).$$

Each bracket is negative because $\kappa_l < 2$, and $\beta_{l,T_l} \neq 0$ for $l \in L'$ (the paper's Lemma 7). Finally $\sum_l a_l\kappa_l(\delta^\star) \leq 3^{1-s}(C_bP)^s\Gamma_s$ as before. For the last sentence, $\gamma(\theta^\star) \geq 3^{1-s}B_p\Gamma_s$. □

Two remarks on what Theorem 19 says. (i) It is pointwise, needs no circularity, and holds at $p = 3$. There the counterexamples violate (I): with constant input components, the mode $\mathbf{1}_T$ coincides with $-\chi_{1,T}$ because $2 \equiv -1 \pmod 3$. This is the non-circularity of the companion note, seen as a spectral coincidence. (ii) It replaces the $\sqrt{|L'|}$ loss of Theorem 17 by nothing. Collective undercutting, and compensation by other fields, are possible *only* through exact arithmetic: an additive relation $\sum \sigma_l\, 2l\mathbf{1}_{T_l} = 0$ among the cheap features, or a mode of some field landing exactly on a cheap feature's sign-flipped character (for instance an input frequency $3l$, whose mode is the conjugate partner of $\chi_{l,T}$). In class $\mathcal{C}$ the second cannot happen, and (I) reduces to the absence of relations $\sum \sigma_l\chi_{l,T_l} \in \{\pm\chi_{k,T}\}$. The programme supports have $K^* = 2, 4, 7$ at $p = 11, 31, 101$ ($b = 2$; exhaustive vertex enumeration at $p = 11$, and at $p = 31$ for supports up to size 4), i.e. $K^* \approx 1.5\ln p$, below the capacity $2\log_3 p \approx 1.8\ln p$ of dissociated sets in $\mathbb{Z}_p^2$. A dissociated choice of T_l exists for the programme supports at $p = 31$ and 211, but none was found at $p = 101$ or 401. Moreover the network, not the proof, decides which T carries a feature's interference. With all interference on $T = \{1, 2\}$ the characters are 1-dimensional, and $\log_3 p \approx 0.9\ln p < K^*$ forces relations, so exact dissociation cannot cover every feature being cheap at large p. There one has to use $\nu = \prod(1-\varepsilon\cos)$ with $\varepsilon < 1$, whose relation terms carry weight $(\varepsilon/2)^{\text{length}}$. The room for that is $1/2$ against the sharp single-feature constant $c_1(2,s) = \sup(|\beta| - a)/\max_T|\beta_T|$, which numerically equals $1 - 3^{1-s}$ (a single neuron; 0.423 at $b = 2$) and so is strictly below $1/2$.

Theorem 20 (the price of a feature at $n = 2$, every prime, arbitrary inputs)**.** *For $n = 2$, $2 \leq b \leq 3$ and every odd prime p, every single-feature network (outputs at frequency 1, inputs arbitrary, any width) satisfies*

$$\min_x \operatorname{cm}(\Phi_1(x)) \;\leq\; \operatorname{Re}\beta_1 - \cos(\pi/p)\max_T|\beta_{1,T}| \;\leq\; 3^{1-s}Q^s \;\leq\; 3^{1-s}B_p\|\theta\|_{2,b}^\nu.$$

Hence $\pi_{2,b}(p) = 3^{1-s} = \nu^{1-s}$: a single feature cannot be bought below the interference-free price at $n = 2$, at any prime. This settles (Q1) for $n = 2$, and with the price–programme theorem the decoupled value is the interference-free value. At $p = 3$ it recovers Part I.

Proof. Fix $T \neq \emptyset$ and $\chi = \chi_{1,T} = 2\mathbf{1}_T$. A mode of Φ_1 at $\pm\chi$ needs input indices k_t with $k_t+1 = \pm 2$ on T and $k_t+1 = 0$ off T. So the only modes there are the sign-flipped mode, with coefficient a, $|a| = |\beta_{1,T}|$, and the conjugate partner (input frequency 3 on T), with coefficient a'; this holds for arbitrary inputs. Put $b = \overline{a'}$ and $c = 1-\rho^{-1} = 2\sin(\pi/p)e^{i\alpha}$, $\alpha = \frac{\pi}{2} - \frac{\pi}{p}$. For $\delta = \pm 1$ the gap $g_\delta = \operatorname{Re}(\Phi_1 c_\delta)$, with $c_{\pm 1} = c, \bar{c}$, has coefficient $\hat{g}_\delta(\chi) = \frac{1}{2}(ac_\delta + b\bar{c}_\delta)$. With $\nu = 1 - \cos(2\pi\langle\chi, x\rangle/p + \arg\hat{g}_\delta(\chi))$,

$h \le \mathbb{E}[\nu g_\delta] = \mathbb{E}g_\delta - |\hat{g}_\delta(\chi)|$. Averaging over $\delta = \pm 1$, $\frac{1}{2}(\mathbb{E}g_1 + \mathbb{E}g_{-1}) = \kappa_{\min} \operatorname{Re}\beta_1$. Moreover writing $ac + b\bar{c} = |c|X$ and $a\bar{c} + bc = |c|Y$, we have $Xe^{i\alpha} - Ye^{-i\alpha} = 2ia \sin 2\alpha$ (the b-terms cancel), so $|ac + b\bar{c}| + |a\bar{c} + bc| \ge 2|a||c||\sin 2\alpha|$. So

$$h \le \kappa_{\min} \operatorname{Re}\beta_1 - \tfrac{1}{2}|a| \cdot 2\sin(\tfrac{\pi}{p})\sin(\tfrac{2\pi}{p}) = \kappa_{\min}\big(\operatorname{Re}\beta_1 - \cos(\tfrac{\pi}{p})|\beta_{1,T}|\big),$$

and $h = \kappa_{\min} \min_x \mathrm{cm}$ (companion note, Lemma 3). Take T maximising $|\beta_{1,T}|$. Part I's Theorem A (valid for arbitrary inputs by the paper's Lemma 6) gives $\operatorname{Re}\beta_1 \le 3^{1-s}Q^s + \frac{1}{2}\max_T |\beta_{1,T}|$, and $\cos(\pi/p) \ge \frac{1}{2}$. Finally $Q = \sum_i |\beta_{i1}|^r \le C_b P$. □

The partner, which is the $p = 3$ mechanism's lever (there $3 \equiv 0$, i.e. the constant input component), cannot help: it can cancel the sign-flipped coefficient at one edge, but not at both, and the residual is the factor $\cos(\pi/p)$. For several features the same averaging over $\pm\delta^\star$ gives the gain $\kappa_l(\delta^\star)|\cos(\pi l\delta^\star/p)|\,|\beta_{l,T_l}|$ per feature when only its own partner interferes, so Theorem 19 survives own partners whenever $|\cos(\pi l\delta^\star/p)| \ge \frac{1}{2}$ for the cheap l.

Every n. The proof of Theorem 19 uses $n = 2$ only through Theorem A. For general n, give each cheap feature l a family $\mathcal{T}_l$ of non-empty $T \subseteq [n]$ with linearly independent indicators (their characters $\chi_{l,T}$ are then dissociated for $p > 2n$), and take the Riesz product over all $\chi_{l,T}$, $T \in \mathcal{T}_l$, $l \in L'$. The same computation gives $\gamma(\theta) \le \nu^{1-s}B_p\Gamma_s(p)$ (strictly if $L' \ne \emptyset$) under (D), (I) for this larger set, provided

$$(\mathrm{H}_n) \qquad |\beta_l| - \nu^{1-s}Q_l^s \;\le\; \tfrac{1}{2}\textstyle\sum_{T\in\mathcal{T}_l}|\beta_{l,T}| \quad \text{for some admissible } \mathcal{T}_l.$$

(H_n) is a single-feature statement about atoms $\beta_i \in \mathbb{C}$, $\xi_i \in \mathbb{T}^n$ (the paper's Lemma 6), i.e. the constant $c_1(n,s) := \sup(|A| - \nu^{1-s}Q^s)/\max_{\mathcal{T}}\sum_{T\in\mathcal{T}}|\beta_T|$ must be at most $\frac{1}{2}$. Local search at $b = 2$ (the largest s) gives $c_1 = 0.211$ ($n = 2$; a single neuron, $\frac{1}{2}(1 - 3^{-1/2})$), ≈ 0.26 ($n = 3$) and ≥ 0.35 ($n = 4$), all below $\frac{1}{2}$ and, with Fejér kernels below, below $\frac{d}{d+1}$. For $n = 2$, (H_2) is Theorem A. For $n \ge 3$ it is the one single-feature inequality still to be proved.

Room in Theorem 19. The factor $1 - \cos\theta$ extracts half of each character's amplitude. A shifted Fejér kernel $F_d(\theta + \pi) = 1 + 2\sum_{k=1}^{d}(1 - \frac{k}{d+1})\cos k(\theta + \pi) \ge 0$ extracts the fraction $\frac{d}{d+1}$. The proof is the same, provided the representations $\sum_l k_l\chi_{l,T_l}$ with $|k_l| \le d$ are unique and $\hat{g}$ vanishes at the multiples $k\chi_{l,T_l}$, $2 \le |k| \le d$. In class $\mathcal{C}$ the latter fails only if some feature is a multiple kl of a cheap feature l. The conclusion then holds whenever $|\beta_l| - a_l \le \frac{d}{d+1}\max_T|\beta_{l,T}|$, which for $n = 2$ leaves a factor $\frac{2d}{d+1}$ of room over Theorem A, and more against the sharp constant $1 - 3^{1-s}$. That room is what a weighted product $\prod(1 - \varepsilon\cos)$ needs to absorb the relations that dissociation cannot exclude at large p. The price is a smaller capacity: unique representations allow at most $n\log_{2d+1} p$ characters.

Proposition 21 (hiding a character needs many modes; any network)**.** *For $\chi \in \mathbb{Z}_p^n \setminus \{0\}$ let $\phi_j(\chi)$ be the coefficient of Φ_j at $\rho^{\langle\chi,x\rangle}$ and $F_\chi(\delta) := \mathbb{E}_x[\sigma(x,\delta)\rho^{-\langle\chi,x\rangle}]$ ($\delta \ne 0$), $F_\chi(0) := 0$. Then $\phi_j(\chi) = -\frac{2}{p}\sum_\delta \rho^{j\delta}F_\chi(\delta)$ for every $j \in F$, and $\phi_j(-\chi) = -\frac{2}{p}\overline{\sum_\delta \rho^{-j\delta}F_\chi(\delta)}$. Consequently, if some field has a mode at $\pm\chi$ and F_χ vanishes on a symmetric set $B \subseteq \mathbb{Z}_p \setminus \{0\}$, then at least $|B| + 1$ of the $p - 1$ coefficients $\{\phi_j(\pm\chi)\}_{j\in F}$ are non-zero.*

Proof. The first formula is the χ-coefficient of the slack identity (Lemma 11); the second follows because σ is real. So $\hat{F}_\chi(j) = \sum_\delta F_\chi(\delta)\rho^{j\delta}$ equals $-\frac{p}{2}\phi_j(\chi)$ for $j \in F$ and $-\frac{p}{2}\overline{\phi_{-j}(-\chi)}$ for $j \in -F$. $F_\chi \not\equiv 0$ and $|\operatorname{spt} F_\chi| \le p - 1 - |B|$, so Tao's uncertainty principle for $\mathbb{Z}_p$ ($|\operatorname{spt} f| + |\operatorname{spt}\hat{f}| \ge p + 1$) gives $|\operatorname{spt}\hat{F}_\chi| \ge |B| + 2$, of which at most one is $j = 0$. □

For $\chi = \chi_{l,T}$ every mode $(j, \pm\chi)$ other than feature l's own needs input energy at an off-frequency ($2l - j$ or $-2l - j$ on T). So hiding one sign-flipped character from the $|B| \geq 2K^*$ binding directions of a programme vertex needs at least $2K^*$ off-frequency mode families. And by the paper's Lemma 6 each such family comes with $2^n - 1$ sign-flipped siblings of equal size at other characters, which must in turn be paid for or hidden. The proposition is exact only where the slack vanishes identically. Turning it into the quantitative statement the compensation case needs (a robust uncertainty principle on $\mathbb{Z}_p$, combined with this sibling conservation) is the remaining analytic step.

The relations case is a one-sided Sidon problem. Suppose all interference sits on $T = \{1, 2\}$, the case in which the network can force relations. Then the conditional gap is $M_\delta + \varphi(w) - \varphi(w - \delta)$ with $\varphi(w) = \sum_l A_l \cos(2\pi l w/p + \theta_l)$ (for two aligned neurons per feature this is exact; see the relations note in `mechanism_scripts`). Evaluating every direction at one common point w_0 and averaging with the programme's dual weights μ (symmetric, on the binding set) gives

$$h \;\leq\; \sum_\delta \mu_\delta M_\delta - \Big(- \min_w \psi_\mu(w) \Big), \qquad \psi_\mu(w) = \sum_l A_l K_l(\mu) \cos(2\pi l w/p + \theta_l),$$

because $\varphi - \mu * \varphi$ has coefficients $A_l K_l(\mu)$. So the interference-free bound for this family follows from the one-sided Sidon inequality $-\min \psi_\mu \geq c \sum_l A_l K_l(\mu)$ with $c = 1 - (2/3)^{s-1}$ (0.18 at $b = 2$), for adversarial phases. That inequality holds with $c = \frac{1}{2}$ for dissociated S (Riesz), and numerically with $c \approx 0.8$ for the programme supports. It fails for arithmetic progressions, whose Rudin–Shapiro polynomials are flat, but those have poor coverage. What is missing is a "coverage implies one-sided Sidon" lemma for $|S| \approx 1.5 \ln p$, which is Chowla's cosine problem with phases and with a coverage hypothesis.

Spurious features need all-slot content. A clean amplitude at q needs a neuron with frequency-q content in every input slot and an output at q. So a cheap single-feature mechanism built from single-frequency inputs, even if it beat the interference-free value, would not change the count: compensating outputs at other frequencies carry no clean amplitude. The count can change only through (i) a shift of the allocation's support or (ii) neurons with multi-frequency inputs in all slots (sign-vector type mechanisms).

Parasitic orbits: compensation is cheap, but does not pay in the models tested. An interference-free block for feature j can carry a hidden compensating mode. Give its three neurons input components at the off-frequency $q = 2l - j$ in slot 1, with relative size ρ and phases $w_i = e^{i \arg \hat{u}_{i1}(q)}$. With the paper's block phases $\alpha_i = \pi(i - 1)/3$, the three sibling modes $(-q, -j), (q, j), (-q, j)$ cancel iff $w_1 = w_2 e^{i\pi/3}$ and $w_3 = w_2 e^{-i\pi/3}$. The surviving mode lands on $\chi_{l,\{1\}} = (2l, 0)$ with size exactly $\rho\, a_j/\sqrt{1 + \rho^2}$, while the block's clean amplitude drops to $a_j/\sqrt{1 + \rho^2}$ at fixed cost. So a compensating mode of size M costs only $O(M^2)$. We tested the most favourable configuration (one feature on two aligned neurons, interference on the single character $(2l, 0)$, all other features carrying parasitic orbits, all phases, sizes and the allocation optimised). In the large-p limit the margin is exactly $\min_\delta [\sum_k a_k \kappa_k/\sqrt{1 + \rho_k^2} + (\frac{3}{2})^{s-1} a_l \kappa_l - |Z(\delta)|]$, where Z is the total coefficient at $(2l, 0)$. Against the exact programme optimum this reaches 0.92 and 0.99 of the IF value at $p = 31, 101$ ($b = 2$); the optimiser always switches the cheap feature off. (An earlier run reported gains; they came entirely from a suboptimal programme solver, since corrected.) By Proposition 21, exact hiding from a programme vertex's $2K^*$ binding directions needs more modes than the features' outputs provide.

Remark 22 (why the remaining step is a question of constants)**.** Let K features each carry interference of size ε (relative to their amplitudes). Their total gain at a binding direction is at most $k^*\varepsilon\sum_l \kappa_l(\delta^\star) \asymp k^*\varepsilon K$. The slack they force is at least $\varepsilon\sqrt{\sum_l \kappa_l(\delta^\star)/(2^n-1)} \asymp \varepsilon\sqrt{K}$ (proof of Theorem 17). This is also the right order for an upper bound: translating each feature's inputs rotates its sign-flipped phases independently, and a sum of K cosines with independent phases has supremum $O(\sqrt{K\log p})$. At the programme's own scale $K = K^* = \Theta(\log p)$ (paper, Theorem 11), gain and forced slack are therefore of the same order, uniformly in p. So collective undercutting is neither negligible nor dominant at any prime, and no order-of-magnitude argument can settle the count. A proof has to compare three sharp constants: the true $k^*(n,s)$ (at least $c_n(s-1)$, from $\nu-1$ equal aligned atoms), the best cancellation constant for the profiles φ_T under the joint pointwise constraint, and the coverage constant of the programme at its binding directions. The experiments at $p = 5, 7$ (no compensated network within $4\times$ of IF) say the first is small against the other two at those primes. Theorem 19 is such a comparison, and it is won by the sharp pair $\frac{1}{2}$ (Riesz gain per feature) against $c_1(2,s) \le \frac{1}{2}$ (Theorem A) whenever the cheap features' characters are dissociated. A dissociated set in $\mathbb{Z}_p^n$ has at most $n\log_3 p$ elements (3^K distinct sums in a group of order p^n). For $n = 2$, $b = 2$ that is $\approx 1.8\ln p$, against $K^* \approx 1.5\ln p$, so exact dissociation is in reach in principle. But an adversarial choice of T (all interference on $T = \{1,2\}$) reduces the capacity to $\log_3 p \approx 0.9\ln p$, which is below K^*. A direct test agrees that phases cannot flatten the profiles: realising every feature of a 6-feature allocation at $p = 31$ by a two-neuron sub-network and optimising all translations gives margin $-0.47\times$ the IF margin, not the > 1 needed. Each feature's field is a product polynomial $P_l(\zeta^l)$, whose coherent excursions phases do not remove.

Statement	Status
$\gamma_b^* = \nu^{1-s} B_p \Gamma_s(p)$ for all p, n, b (Conjecture 20)	**false**: $p = 3$, $b = 2$, $n \geq 11$; excess 2.21× certified at $n = 18$, growing
$\nu^{1-s} B_p \Gamma_s \leq \pi_{n,b}(p) B_p \Gamma_s \leq \gamma_b^* \leq B_p \Gamma_s$	proved (companion note)
$\gamma_b^* = \pi_{n,b}(3)\, B_3 \Gamma_s(3)$ at $p = 3$	proved (one frequency)
width $\geq \nu - \lfloor \nu/p \rfloor$ for shared-sign networks	proved (Theorem 3)
single-frequency networks cannot beat IF when $k_p \geq k^*(n, s)$	proved (companion note); k^* numerical
$\gamma_b^* = \pi_{n,b}(p) B_p \Gamma_s(p)$ (Conjecture 20′, decoupling)	open; genuine obstacle exhibited (Section 6)
$\pi_{n,b}(p) = \nu^{1-s}$ for $p \geq 5$	open; every mechanism tested loses, by a factor growing in n
feature count of every global optimiser = $K^*(s, p)$, $p = 3$	proved (Prop. 6(a))
feature count = $K^*(s, p)$, all p, dense range of Thm. 8	proved (Thm. 8; extends $[b_0, \nu]$ in all cases computed)
feature count = $K^*(s, p)$, all p, given decoupling	proved (Prop. 7); price irrelevant
feature count = $\Theta(\log p) = \Theta(K^*)$, all p, unconditional	proved (Thm. 9 with paper's Cor. 12)
single-frequency networks, $n = 2$, all $p \geq 5$, $b \in [2, 3]$: no gain over IF	proved (Cor. 10)
$\lvert\beta_j\rvert \leq \nu^{1-s} Q_j^s + K_{\nu,s} \ell_j$ (Lemma 7 with interference), every n	proved (Lemma 13)
single-frequency networks, every n, b, all $p \geq p_0(n, b)$: no gain over IF	proved (Cor. 14)
count = K^* in class $\mathcal{C}$, $n = 2$, all p, $b \in [2, 3]$, ≤ 2 features below IF price	proved (Thm. 17)
count = K^* for every maximiser with no feature bought below IF price, all n, p, b	proved (Prop. 18)
$n = 2$, any network: cheap features force additive relations or spectral coincidences	proved (Thm. 19)
$\pi_{2,b}(p) = 3^{1-s}$ for every prime p (single features, arbitrary inputs): (Q1) at $n = 2$	proved (Thm. 20)
feature count = $K^*(s, p)$, $p \geq 5$, $b < b_0$	open; reduced to relations among cheap features and off-frequency compensation (Sec. 8)

Answer to "prove that the unrestricted optimum matches the prediction". In the interference-free form this cannot be done: the statement is false at $p = 3$, and increasingly so as n grows. In the price–programme form it holds at $p = 3$ by definition, and there the object to compute is $\pi_{n,b}(3)$. We have shown that, in the class that contains every counterexample, this reduces to choosing $\nu - \lfloor \nu/3 \rfloor$ real slopes, with the weights given by an LP. For $p \geq 5$ the full statement would follow from two claims:

(Q1) $\pi_{n,b}(p) = \nu^{1-s}$, i.e. a single feature cannot be bought below the IF price. Theorem 3 limits the width saving of the $p = 3$ mechanism to $\lfloor \nu/p \rfloor$ neurons, and to none for $p > \nu$. The circularity theorem of the companion note handles single-frequency inputs when $k_p \geq k^*$. A proof needs a cost-weighted (robust) form of the width theorem for arbitrary sign patterns.

(Q2) Decoupling. It has to control compensation by interference-only fields, which demonstrably raises a feature's margin.

Both hold in all our experiments at $p = 5, 7$, by margins of $4\times$ or more that grow with n.

Files. `certify_1d.py` (rigorous certificate for the 1-D family, any p), `oned_certified/` (parameters of the networks in Table 1), `mechanism_scripts/` (the field model, the LP, the slope search and continuation, the joint two-frequency LP, the release test).